Team Lighthouse
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ODP 6: Client Report

Context and Problem Statement:

Spotted lantern flies (SLFs) are currently impacting grape harvests massively, almost 300 million dollars are lost annually in the grape and wine industry in New York State alone. On top of affecting yield, they also impact the quality of the harvests. If we can limit their destruction by attracting the spotted lantern flies to hotspots away from the fields, we can save farmers thousands of dollars and reduce food waste.

Final Prototype and Application:

We have built a mockup of the mechanical mechanism for our diffuser system which would store chemicals such as Tree of Heaven sap, honeydew, or different combinations of these compounds. An anemometer is attached to the top that has been linked to a fan below the lid by a gear system to induce airflow within the box when wind flows. The goal is to mimic a more natural release pattern. We have also incorporated a mounting system which involves ratchet strapping the box to any structure needed e.g trees, telephone poles etc. However, future iterations of the design may be given modular attachment methods to allow for easier application though that will not be a major priority informing future design direction.

This will be relevant for our clients as they have the ability to choose a hotspot for the spotted lanternflies rather than being controlled by the affected harvest. They would be able to use this with the intuitive mounting system and be able to expand it by simplifying getting more of these lures. As chemicals get developed to attract these spotted lanternflies more efficiently, the liquid can easily be replaced and the mechanism still remains the same.

Conclusion and Recommendation:

From our efforts to solve this issue, we determined that the optimal next steps should be to get in contact with growers and test our solution where it is needed. In this way, we could obtain more detailed data on the application sites for our product and the conditions under which it must operate. Some considerations that we have in preparation for testing are determining how we can make the system function at lower wind speeds (such as by modifying in favor of lighter components or using smoother edged parts to reduce friction) and how we can deal with the SLFs after luring them. Based on our test results, our main issue is the high wind speed that must be reached before the system starts given that the system moved minimally during testing. Thus, going into the field will be tremendously helpful for informing our future redesigns as we will be able to know immediately how effective a new design proposal will be. For future prototypes, we are thinking that it is important to prioritize rotational speed for the turbine so that the fans within the box will also spin faster. In order to accomplish this goal, we plan to experiment with more efficient turbine designs that will optimize wind speed by looking at how modern wind turbines are designed. However, as the design is meant to be used for long-term applications we want the system to

release the pheromones at a steady and constant rate. Hence, it is important that the rotational speed is not too high.

Testing Results:

Below is the volumetric flow rate comparison that was conducted based on our final testing with Gears attached.

$P_{wind} = \frac{1}{2}\rho \times A_{anemo} \times v_{wind}^{3}$ - wind power captured by anemometer

$P_{jet} = \frac{1}{2}\rho \times A_{slit} \times v_{slit}^{3}$

$v_{slit} = 0.447 m/s$

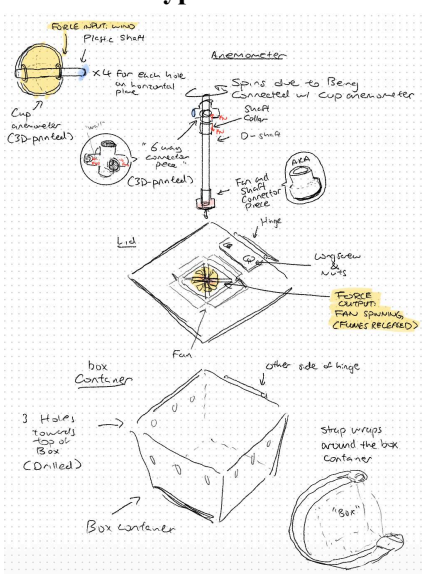
$v_{anem} = 4.92 m/s$

$\eta = \frac{0.00196 \times v_{slit}^{3}}{0.000931 \times v_{wind}^{3}} = 0.15\%$

This may be a small efficiency but it is a more accurate representation of the efficiency than what we used for the initial testing. The initial testing is in the next section. This low efficiency result showcases the additional amounts of friction etc involved in the new design.

Prototype and Testing Details:

Initial Prototype



Initial Tests

Stability of Anemometer Assembly (6 way connector + Cup Anemometers):

Height of Anemometer vs Produced Wind Speed (measured by a digital wind speed indicator)

Results:

Constant Applied Wind Speed: 5.1 mph

Height (in)	Trial 1 (mph)	Trial 2 (mph)	Trial 3 (mph)	Stability
4	0.6	0.7	0.7	Most Stable
6	0.6	0.6	0.7	More Stable

10 (Original)	0.6	0.6	0.6	Pretty Shaky

Iteration for next prototype:

We will lower the anemometer mechanism assembly and shorten the main D-Shaft to cut down on the weight, and increase stability by lowering the system's center of mass to the container. Our current assembly and test design isn't quite sensitive enough to show a clear improvement (baseline measurement of our air speed meter is 0.6 mph, and only measures in one decimal increments), but we observed the anemometer to spin much quicker at lower heights.

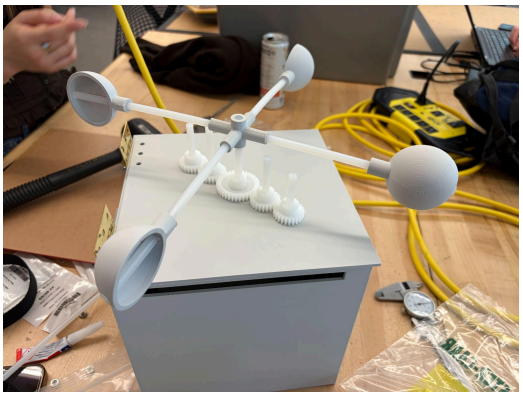
Applied Wind Speed vs Produced Wind Speed (measured at the slits in the box)
Results:

Applied Wind Speed (mph)	Produced Wind Speed (mph)	Efficiency(%)
5.1	0.7	14
3.1	0.6	19
4.0	0.6	15
4.6	0.7	15
3.8	0.6	16
4.4	0.7	16

Iteration for next prototype:

We would add a gear ratio into the anemometer/fan linkage so the energy produced by the anemometer is amplified instead of having it directly spin the fan, since our efficiency is fairly low at the moment. Again, our test system is not quite sensitive enough to capture distinctions between the test results, so, hopefully, we will be able to capture more of a difference with higher wind speeds produced by using higher precision equipment in the future.

Second Prototype



Iteration for next prototype:
We would add a bigger gear ratio into the anemometer/fan linkage. The gear ratio did work but the limitation of McMaster was that the biggest ratio was 1.5 when a bigger one would help a lot. We would also experiment with the materials and designs used in place of the cups in the anemometer design so that it would maximize the input air force while minimizing the weight.

References

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Bill of Materials

Everything needed in the Final Product (The Component List of the final prototype)

Part	Part #	Quantity	Description	Cost
Cinching Strap with Squeeze-Release Buckle, Nylon, Black, 4ft Long	29695T991	2	1" Wide	$11.14
Ultra-Low-Friction UHMW Polyethylene Bars, 1/4", 3ft long	8701K37	1	1/4" Diameter	$4.68
Clamping Shaft Collar for 1/4" Diameter, 303 Stainless Steel	6435K32	2	1/4" Diameter	$16.40
Ultra-Low-Friction UHMW Polyethylene Bars, 1/4", 3ft long	8701K37	2	1/4" Diameter	$9.36
20 Degree Pressure Angle Plastic Gear, Round Bore, 24 Pitch, 36 Teeth	2662N11	2	Round Bore, 32 Pitch, 36 Teeth	$6.77
20 Degree Pressure Angle Plastic Gear, Round Bore, 24 Pitch, 24 Teeth	2662N9	4	Round Bore, 32 Pitch, 48 Teeth	$22.76
Ball Bearing, Open, Trade Number R168, for 1/4" Shaft Diameter	57155K355	5	1/4" Shaft Diameter	$30.00
Surface-Mount Hinge with Holes, Brass, Nonremovable Pin, 2" x 11/16" Door Leaf	1603A27	2	2" x 11/16" Door Leaf	$6.96
Box Case + Lid		220 g	PLA @ 0.05	11
Six Way Connector Piece		5 g	PLA @ 0.05	0.25
4 Cups		20 g	PLA @ 0.05	1
Box Case + Lid		875 g	PLA @ 0.05	43.75

Gemfan D90 90mm 3.5" Ducted 3-Blade Propeller - M5 Shaft (Set of 4) - 5mm - Black		1	3.5" Ducted 3-Blade Propeller - M5 Shaft- 5mm	$3.49

Bill of Materials (Overall Process)

Part	Part #	Quantity	Description	Cost
DC Equipment Cooling Fan	8774N26	3	1.97" Square Hole, .39" Deep	$14.49
Cinching Strap with Squeeze-Release Buckle, Nylon, Black, 4ft Long	29695T991	2	1" Wide	$11.14
Quick-Tight Tie Down Strap, 3ft Long	8834T651	1	2" Wide	$15.20
PTFE Shim Stock	1192N11	3	8"x12" Sheet, 0.001" Thick	$11.01
D-Profile Rotary Shaft, 1045 Carbon Steel	8632T139	1	12" Long, 1/4" Diameter	$10.86
Ultra-Low-Friction UHMW Polyethylene Bars, 1/4", 3ft long	8701K37	1	1/4" Diameter	$4.68
Plastic Hinges without Holes, Clear Acrylic Plastic	11185A141	10	1/2" x 11/16" Door Leaf	$8.80
Clamping Shaft Collar for 1/4" Diameter, 303 Stainless Steel	6435K32	2	1/4" Diameter	$16.40
Ultra-Low-Friction UHMW Polyethylene Bars, 1/4", 3ft long	8701K37	1	1/4" Diameter	$4.68
20 Degree Pressure Angle Plastic Gear, Round Bore, 24 Pitch, 36 Teeth	2662N11	2	Round Bore, 32 Pitch, 36 Teeth	$6.77

20 Degree Pressure Angle Plastic Gear, Round Bore, 24 Pitch, 24 Teeth	2662N9	4	Round Bore, 32 Pitch, 48 Teeth	$22.76
Ball Bearing, Open, Trade Number R168, for 1/4" Shaft Diameter	57155K355	5	1/4" Shaft Diameter	$30.00
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Reflection (Andy)

I went to a conference during the final in-person presentation. I planned around this by informing my group ahead of time and working with them in-person and over text beforehand to talk about which points we should emphasize for the presentation. I made sure to stay updated on the progress of the actual presentation and tried to answer whatever questions were asked and discussed. The team made whatever adjustments last minute in person but involved me during the planning process throughout the days beforehand. In the future, for teams facing a similar absence, I would recommend communicating the details of the absence far in advance so the team has plenty of time to plan around this while keeping the absentee involved.